

Due: Thursday April 7th, latest at 2PM

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1. We consider a magnetostatic situation, i.e., currents and fields are time-independent.

- (a) [2 pts] The vector potential in a region is given by

$$\mathbf{A}_1 = \left(-\lambda \frac{z}{2}\right) \hat{j} + \left(\lambda \frac{y}{2}\right) \hat{k}$$

Find the magnetic field $\mathbf{B}$. Which direction does the magnetic field point toward?

- (b) [1 pt] Use Ampere's law in differential form to find the current density $\mathbf{J}$ in this region.

- (c) [2 pts] Now consider the vector potentials

$$\mathbf{A}_2 = (4\lambda) \hat{i} + (\lambda y) \hat{k}, \quad \mathbf{A}_3 = \left(\lambda \frac{y-z}{2}\right) \hat{j} + \left(\lambda \frac{y+z}{2}\right) \hat{k}$$

Show that both these vector potentials lead to exactly the same magnetic field as the vector potential $\mathbf{A}_1$.

- (d) [2 pts] If two choices of the vector potential correspond to the same magnetic field, their difference must have zero curl. Write down the vector function $\mathbf{A}_2 - \mathbf{A}_1$ and show that it indeed has zero curl.

- (e) [2 pts] A vector with zero curl can be written as a gradient of some scalar function; thus $\mathbf{A}_2 - \mathbf{A}_1 = \nabla f$. By examining the form of $\mathbf{A}_2 - \mathbf{A}_1$, guess a scalar function f that does this job. Show that the gradient of your function indeed gives $\mathbf{A}_2 - \mathbf{A}_1$.

- (f) [2 pts] Does $\mathbf{A}_1$ satisfy the Coulomb condition (or Coulomb gauge)? Does $\mathbf{A}_3$ satisfy the Coulomb condition?

2. Steady current I flows through an infinitely long straight wire placed along the z axis, from $z = -\infty$ to $z = +\infty$ through the origin.

We will use cylindrical coordinates, (r, ϕ, z) , in this problem. The unit vectors are denoted as $\hat{r}$, $\hat{\phi}$, $\hat{z}$. You will need to recall or learn what these coordinates and unit vectors mean. Note that the directions of the $\hat{r}$ and $\hat{\phi}$ vectors depend on the angle ϕ .

- (a) **[3 pts]** Write down an expression for the magnetic field vector $\mathbf{B}$ created by the current I at point (r, ϕ, z) . This should be a vector equation. What are the components B_r , B_ϕ and B_z ?
- (b) **[5 pts]** Use your expression for $\mathbf{B}$ to calculate the vector potential $\mathbf{A}$. You might need to recall that the curl of a vector $\mathbf{v}$ in cylindrical coordinates is

$$\nabla \times \mathbf{v} = \left(\frac{1}{r} \frac{\partial v_z}{\partial \phi} - \frac{\partial v_\phi}{\partial z} \right) \hat{r} + \left(\frac{\partial v_r}{\partial z} - \frac{\partial v_z}{\partial r} \right) \hat{\phi} + \frac{1}{r} \left(\frac{\partial(rv_\phi)}{\partial r} - \frac{\partial v_r}{\partial \phi} \right) \hat{z}$$

Also note: the symmetry of the problem ensures that $\mathbf{A}$ has no z -dependence.

3. We consider now a *thick* wire, of circular cross-section, with radius R . The wire is infinitely long, and its axis is placed along the z axis. Steady current I flows through the wire from $z = -\infty$ to $z = +\infty$. We will continue to use cylindrical coordinates.

First consider the current I to be uniformly distributed over the circular cross-section, so that the current density is

$$\mathbf{J} = \left(\frac{I}{\pi R^2} \right) \hat{z} \quad \text{for } r \leq R \quad \text{and} \quad \mathbf{J} = 0 \quad \text{for } r > R$$

In class, we used Amperian loops to show that the magnetic field is

$$\mathbf{B} = \left(\frac{\mu_0 I}{2\pi} \frac{r}{R^2} \right) \hat{\phi} \quad \text{for } r \leq R \quad \text{and} \quad \mathbf{B} = \left(\frac{\mu_0 I}{2\pi} \frac{1}{r} \right) \hat{\phi} \quad \text{for } r > R$$

- (a) **[4 pts]** Show that Ampere's law in differential form ($\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$) is satisfied outside the wire ($r > R$). Use cylindrical coordinates.
- (b) **[4 pts]** Show that Ampere's law in differential form is satisfied inside the wire ($r < R$). Use cylindrical coordinates.
- (c) **[4 pts]** Express $\mathbf{B}$ inside the wire ($r < R$) in Cartesian coordinates. A drawing might help you to find the Cartesian components of $\mathbf{B}$, or you can do this algebraically.
- (d) **[2 pts]** Using Cartesian coordinates, show that Ampere's law in differential form is satisfied inside the wire ($r < R$).

4. We continue with the thick-wire geometry of the previous problem.

Now imagine that the current density is larger near the outer region of the wire cross-section than at the center: $J = kr$, so that $I = \left(\frac{2}{3}\pi R^3\right)k$. (You might want to derive this for yourself before working on the problems below. You may have done something similar for problem set 04.) We thus have

$$\mathbf{J} = kr\hat{z} = \left(\frac{I}{\frac{2}{3}\pi R^3}r\right)\hat{z} \quad \text{for } r \leq R \quad \text{and} \quad \mathbf{J} = 0 \quad \text{for } r > R$$

- (a) [4 pts] Construct an Amperian loop and use Ampere's law in integral form, to find the magnetic field outside the wire ($r > R$). Express $\mathbf{B}$ in cylindrical coordinates. Express your answer first in terms of I , and then in terms of k .
- (b) [5 pts] Use an Amperian loop and Ampere's law in integral form, to find the magnetic field inside the wire ($r < R$). Express $\mathbf{B}$ in cylindrical coordinates. Express your answer in terms of I and in terms of k .
- (c) [3 pts] Show that your expression for $\mathbf{B}$ inside the wire ($r < R$) satisfies the differential form of Ampere's law.

5. Divergence of curl:

- (a) [2 pts] Using Cartesian coordinates, show that the divergence of the curl of any vector $\mathbf{v}$ is zero.
- (b) [3 pts] Maxwell's fourth equation is

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

The divergence of the left hand side is zero; therefore the right hand side also needs to have zero divergence. Show that the divergence of the right hand side is indeed zero. You will need to use the continuity equation and Maxwell's first equation. Point out clearly the step where the continuity equation is used, and the step where Maxwell's first equation is used.